

A Hybrid Approach of Logistic Regression with Grid Search Optimization in Credit Scoring Modeling for Financial Institutions

Sun Thurain Moe
University of Computer Studies
Yangon, Myanmar
sunthurainmoe@gmail.com

Than Than Nwe
University of Information Technology
Yangon, Myanmar
thanthannwe.cu@gmail.com

Abstract— With the rapid development of data science and technology, the credit scoring system has changed with many improvements. Building and perfecting a credit scoring system is critical to economic development and economic system reform. On the contrary, the gap of the brand-new personal system facilitates financial institutions like microfinances and banks to perform additional credit functions for various groups of individuals. The purpose of this study was to develop and test an applied credit scoring model that would assist financial institutions in evaluating credit claims. Given the growing importance of credit to our society and concerns about risking it in the wake of the recent financial turmoil, a robust credit rating model will strengthen confidence in the assessments of financial institutions. There is already a stream of literature associated with credit rating. However, this paper emphasizes the need to demonstrate the effectiveness of discriminating between good and bad customers by applying data mining techniques.

Keywords—credit scoring, retail lending, unsecured loans, credit risk, credit rules engine

I. INTRODUCTION

In the form of financial retail, such as credit cards, credit score directly affects the profitability of financial institutions and is a key analysis. Increasing the demand of credit scoring system requires by the financial institutions are to develop machine-driven risk assessment systems (scoring) that can perform customer assessments and complex product evaluations (ratings) to rapidly assess business operations and measure repricing rates.

A credit scoring is a grouping of completely different variables. It is a method of analysis and classification. It is predicated on a comparison between new and old credit customers. It is a system that automatically evaluates the credit performance of customers with good or bad credit activities and provides the requested credit approval or rejection with high accuracy.

Credit scoring is the most common way to assign credit limit to the specific customers in the retail lending such as credit cards and mortgage loans. Based on Fair Isaac Company (2015) data, more than 75% of credit institutions use a credit scoring for mortgages and more than 90% use it for credit cards [1]. In this way, an accurate assessment of the probability of default can help the banks rate their customers and improve the efficiency of credit relief processes. In addition, a little improvement in estimating the probability of default can bring substantial benefits to the financial institutions such as microfinances and banks.

To summarize, the purpose of credit scoring is to classify loan applicants into two categories: those with good credit who are more likely to repay their financial obligations and those with bad credit who are more likely to default [2].

This paper has the objective of identifying and improvement the ratios better discriminate the credit customers as good or bad customers and to know what is the relative importance of these ratios. In the next section the methodology, techniques of analysis and variables are identified. Finally train the credit scoring model and the empirical results are analyzed.

II. PREPROCESSING

A. Data Cleaning

The dataset is available on Kaggle, and it is collected the customers' demographic data of the consumer loans issued by the Lending Club, a US P2P lender. The classification goal is to predict whether the client as good or bad and subscribe as (0/1) to a term of dependent variable y . The dataset contains more than 450,000 consumer loans data issued between 2007 and 2014 with 75 attributes, including current loan status, customers' demographic data and their payment behavior. Initial data preprocessing does the following:

- Drop all features more than 80% of missing values since the high proportion of missing values may produce false positive results.
- Drop all features not related to credit risk, e.g., id, member_id, url, title.
- Drop all features to be filled only once the customer has defaulted, e.g., recoveries, collection_recovery_fee.

B. WoE and IV Feature Engineering

Binning for all the numeric variables based on Weight of Evidence (WoE) and Information Value (IV) has been widely used in the credit scoring domain. WoE and IV feature engineering are the most important and time-consuming steps to do before developing a credit risk model.

By conducting WOE and IV analysis, the following points can be made accordingly.

- The independent of each variable on the class label can be determined.
- The relationship between class labels and variable can be classified as linear relationship or non-linear relationship.
- The predictive power of the variables can be estimated and the rank of the variables can be determined by that factor.
- Correlations between class labels and variables can be visualized.
- Continuous variables can be used to replace with dummy variables.

- Predict power of missing values can be estimated.

WoE estimates the predictive power of the independent variable relative to the dependent variable. It is generally described as a measure of separating good customers from bad customers by count. Bad customer refers to the customers who defaulted on a loan and good customer refers to the customers who paid back loan regularly.

$$WoE = \% \text{ of Good} / \% \text{ of Bad} \quad (1)$$

IV is one of the most useful techniques to select important variables in a predictive model. It calculates the value to rank variables on the basis of their weight of evidence. The IV is calculated using the following formula and selected the feature according to the rules shown in Table I.

$$IV = \sum (\% \text{ of Good} - \% \text{ of Bad}) * WoE \quad (2)$$

After separating the feature columns containing numeric values into bins with WoE and IV, the resulting bins are combined based on the following factors.

- The number of samples included in each bin must be at least 5% of the total number of training samples.
- Training samples containing missing values must be divided into separate bin.
- The class labels of the training samples included in each bin must have both good and bad.
- Bins with similar WoE values must be combined into one.
- The WoE value of each bin must not be too large or too small.

The above factors are described in the book, Credit Scoring and its Applications [5], published in January 2002.

For the IV, the IV value of each bin must be between 0.2 $\leq$ IV $\leq$ 0.5. Bin with IV value that are too small or too large are combined with bin that are close to the WoE value.

TABLE I. RULES RELATED TO INFORMATION VALUE

Information value	Predictive power
< 0.02	Useless
0.02 - 0.1	Weak
0.1 - 0.3	Medium
0.3 - 0.5	Strong
> 0.5	Suspiciously good

C. One-Hot Encoding

Categorical variables are generally do not have a normal relationship with dependent variable y and the integer encoding is not enough most of the time. In fact, allowing the model to assume a natural ordering between different categories may lead to poor performance or leading unexpected results. Generally, most machine learning algorithms, apart from the decision tree algorithms, are not comfortable working with categorical data. Therefore, categorical data needs to be converted into numerical form. In this study, we need to convert all of our data into a binary format that contains only 1's and 0's. So that all the values of

categorical variables must be convert into binary. This mean that variables with only two values, one and zero and all the categorical variables are defined as the dummy variables. This process sometime called binarization.

The easiest way to change that is with an Integer Encoding algorithm that replaces unique category values with integer values. For example if it is a male it is replaced with 1 and if it female replaced with 0. However, doing so is not always convenient. This is because it is difficult to predict how within category values are related to each other. Another thing is that binary values have only two 1s and 0s, so category values with more than two cannot be converted with Integer Encoding. Therefore, we chose a One-Hot encoding algorithm that uses dummy variables.

The way the One-Hot encoding algorithm works is to create new dummy variables by combining variable names and category values. Then, the relevant category values are encoded as 1 under the corresponding dummy variable and 0 under the remaining dummy variables. The disadvantage of this method is that it increases the number of new variables and reduces the performance of most of the machine learning algorithms. Because of this, it is usually necessary to do features selection after one-hot encoding is done.

III. RECURSIVE FEATURE ELIMINATION (RFE)

Because the categorical data in our dataset were created as dummy variables using the One-Hot Encoding algorithm to convert them into binary format, the dimensions were greatly increased. An increase in dimensions reduces the performance of machine learning algorithms and will leads to overfitting. To overcome that problem and reduce the number of features, we chose the RFE algorithm.

Recursive Feature Elimination (RFE) is a method that fits a model and removes weak feature (or features) until a specified number of features is reached. Certain number of features are required to maintain an RFE, but it is not known in advance how many features are valid. Cross-validation is used to find the optimal number of features along with RFE to score different feature subsets and select the best scoring feature set [3].

There are two important configuration options when using RFE. First, a choice in the number of features to select, and second, a choice in the algorithm used to help select features. Both of these hyperparameters can be explored, although the performance of the method does not strongly depend on these hyperparameters being well-tuned. RFE is wrapper-type and different machine learning algorithm is used to help select features. We used Logistics Regression algorithm as a core of RFE.

IV. GRID SEARCH

The same machine learning model uses different parameters to generalize different data patterns, such as weights or learning rates. These parameters are called hyperparameters. The model must then be iteratively tuned to best solve the machine learning problem. Hyperparameter optimization finds a set of hyperparameters that produce the best model that minimizes a predefined loss function on given discrete data. The objective function takes a number of hyperparameters and returns the correlation loss. Cross-validation is often used to estimate this generalizability. The result of the grid search is the optimal combination of one or more hyper parameters that provides the best model that

accommodates the bias variance transition. This is because cross-validation techniques are used to train and test the model and combine the optimal parameters.

V. LOGISTIC REGRESSION

Logistic regression is the measuring of relationship between the dependent variable (the target) and one or more independent variables (the features). In other words, it is estimating the probabilities with the help of its underlying logistic function. The logistic regression model calculates the sum of the input features and the logistic of the output. The output of logistic regression is converted into probability always between 0 and 1. The higher the value, the probability of current sample is classified as class=1, and vice versa.

$$\text{Logit}(P_i) = 1 / (1 + \exp(-P_i)) \quad (3)$$

$$\ln(P_i / (1 - P_i)) = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n \quad (4)$$

$$P(X) = 1 / (1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}) \quad (5)$$

Where $X = (x_1, x_2, \dots, x_n)$ represents the feature set and $(\beta_1, \beta_2, \dots, \beta_n)$ are coefficients while β_0 is intercept. P represents the probability that is predicted to be bad.

VI. PROPOSED SYSTEM

The term credit scoring is used to describe the process of evaluating the risk a customer poses of defaulting on a financial obligation [4]. The goal is to assign customers to one of two groups: good and bad. A member of a good team is considered likely to repay their financial obligations. A member of a bad group is considered likely to default on their financial obligations. In its simplest incarnation, a credit scorecard consists of a set of characteristics used to assign a customer a credit score that indicates their level of risk.

In this work, four major activities are carried out the whole process of credit scorecard modeling. Data pre-processing is first and the most important step that includes treatment for missing values and drop all the irrelevant features from credit scoring. Then the dataset is split up into 70% of training and 30% of testing set.

As a second, calculate WoE and IV values for all numerical features to bin continuous values. It is the most critical step in credit scoring domain. The rules related to binning numerical features are:

- Each and every bin should not less than 5% of dataset and should have both Good or Bad class labels.
- The bins that have equal or nearest WoE values should be aggregated together.
- All the feature with very low or very high IV value should be dropped.

Fig. 1, 2 and 3 show the binning result of loan_amnt by WoE and IV feature analysis. Fig. 4 and 5 shown the binning result of open_acc variable.

Then create dummy variables as a third for all categorical and binned numeric features. This means that all the features are transformed into binary features and have only two values, zero or one. After that we applied the feature selection method RFE to select 150 features from 202 total dummy features shown in Fig. 6.

Finally fitting the logistic regression model on the training dataset and optimized the hyperparameters with grid search method. Grid search method control the scoring model to force more penalty on false negative than on false positive in highly imbalanced dataset, which is usually important in credit scoring. It will take couple of minutes to an hour based on the training data size.

	Bin	Count	Count (%)	Non-event	Event	Event rate	WoE	IV	JS
0	(-inf, 7012.50]	98080	0.210343	10109	87971	0.896931	-0.065737	0.000886	0.000111
1	(7012.50, 8012.50]	27246	0.058432	2913	24333	0.893085	-0.024806	0.000036	0.000004
2	(8012.50, 15037.50]	164863	0.353567	17650	147213	0.892941	-0.023301	0.000190	0.000024
3	(15037.50, 17037.50]	26178	0.056142	3115	23063	0.881007	0.095844	0.000535	0.000067
4	(17037.50, 19887.50]	29127	0.062466	3675	25452	0.873828	0.162603	0.001759	0.000220
5	(19887.50, inf)	120791	0.259050	13506	107285	0.888187	0.025489	0.000170	0.000021
6	Special	0	0.000000	0	0	0.000000	0.0	0.000000	0.000000
7	Missing	0	0.000000	0	0	0.000000	0.0	0.000000	0.000000
Totals		466285	1.000000	50968	415317	0.890693		0.003577	0.000447

Fig. 1. WoE and IV analysis of loan_amnt

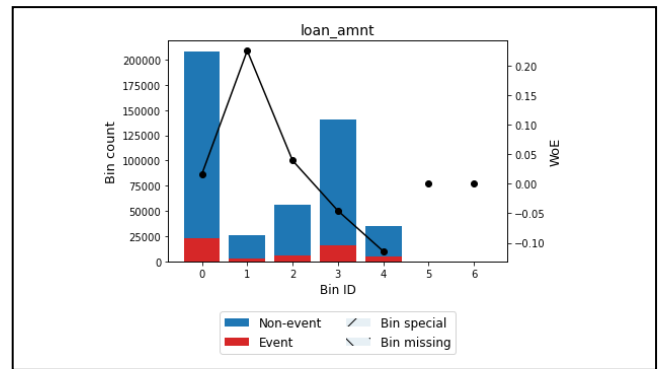


Fig. 2. WoE binning metric loan_amnt

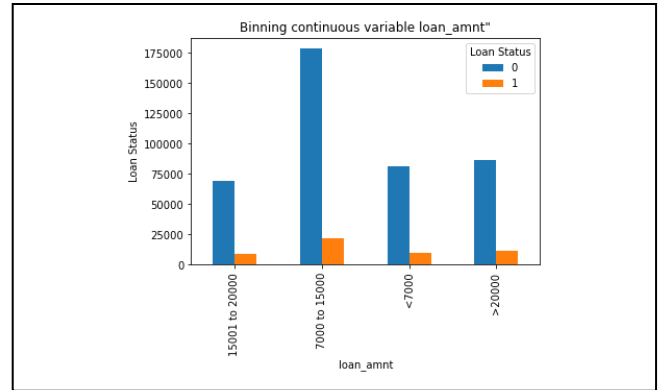


Fig. 3. Binning continuous variable loan_amnt

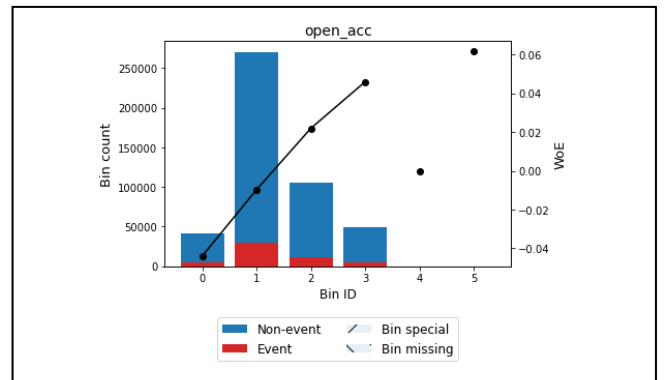


Fig. 4. WoE binning metric open_acc

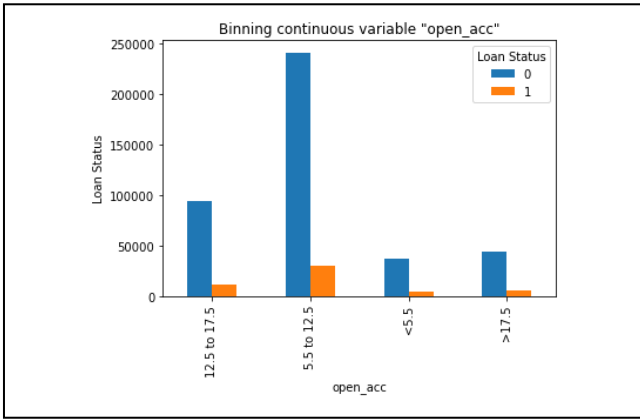


Fig. 5. Binning continuous variable open_acc

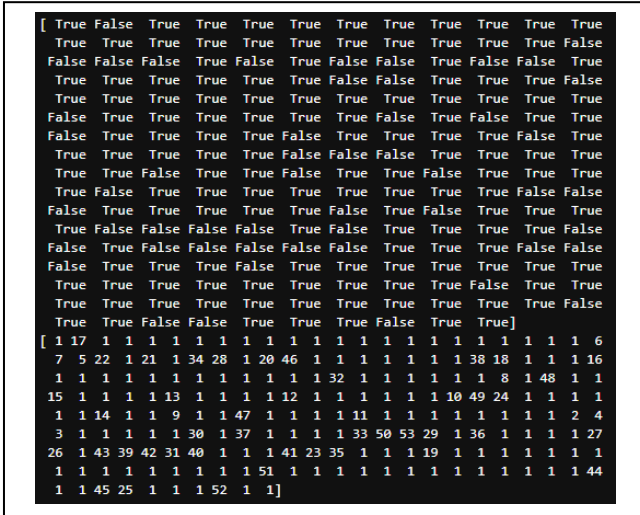


Fig. 6. Feature selection by RFE

VII. EXPERIMENTS AND RESULTS

Credit scoring dataset is randomly split into two separate sets, training (70%) and testing (30%). Then K-Fold testing perform repeatedly in the training set to preliminarily evaluate the model while the test set remain untouched until the final model evaluation.

As per the credit scoring model usually focus on the bad recall rate in order to reduce Non-Performing Loan (NPL) rate of loan portfolio, the final evaluation result with the testing dataset shows that our credit scoring model is achieved higher approval rate and also efficient in true positive rate to reject bad customers.

Our model gives the results 0.91 of good customers recall, 0.86 of bad customers recall and 0.9 overall accuracy. It means that our model is higher approval rate in loan origination and will lower NPL rate in total loan portfolio. Moreover, the result of Area Under Curve (AUC) value 0.88 and the p-value 0.78 of Kolmogorov-Smirnov (KS) test also confirmed the factor mentioned above and considered the model as acceptable and also applicable.

We tested our model on a total of 139,886 loan records. Among them, there are 124,576 good customers and 15,310 bad customers. Out of 124,576 good customers, our model was able to correctly classify 113,275 as good and incorrectly classify 11,301 as bad. Similarly, out of 15,310 bad customers, 13,113 were correctly classified as bad and 2,197 were incorrectly classified as good. If we look at the result as

a percentage, we were able to correctly classify 91% of the good customers and 86% of the bad customers. In terms of accuracy, our model's ability to classify correctly is 90%. In terms of approval rate, out of a total of 139,886 customers, 115,472 were approved for loans. It is 82% of total customers.

A credit scoring model can't just correctly classify the good customers but also how many bad customers can be classified correctly is a key. This is because as more and more bad customers are granted for loans, the percentage of financial loss increases drastically. Because the amount of money can lose from one bad customer is usually more than the total interest rate will get from at least 5 good customers.

Therefore, when looking at the results of a credit scoring model, the first thing to pay attention how good the bad recall rate is. Now our model has only 2197 bad customers out of 115,472 people granted for loans. In terms of percentage, it is only 1.9%, and the NPL rate of a loan portfolio will be well maintained. Detail experimental results shown in the following Fig. 7, 8, 9, 10 and Table II.

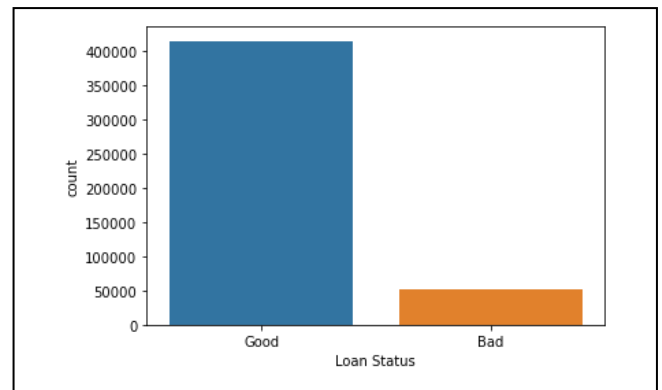


Fig. 7. Class label distribution

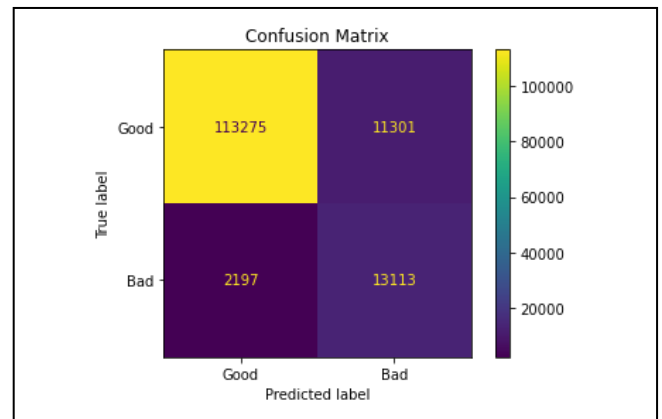


Fig. 8. Confusion matrix

TABLE II. PRECISION, RECALL AND F1-SCORE

	Precision	Recall	F1-Score	Support
Good (0)	0.98	0.91	0.94	124576
Bad (1)	0.54	0.86	0.66	15310
Accuracy			0.9	139886
Macro avg	0.76	0.88	0.8	139886
Weighted avg	0.93	0.9	0.91	139886

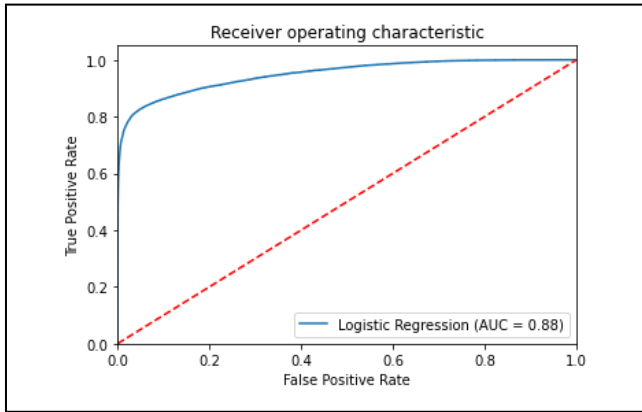


Fig. 9. Receiver operating characteristic

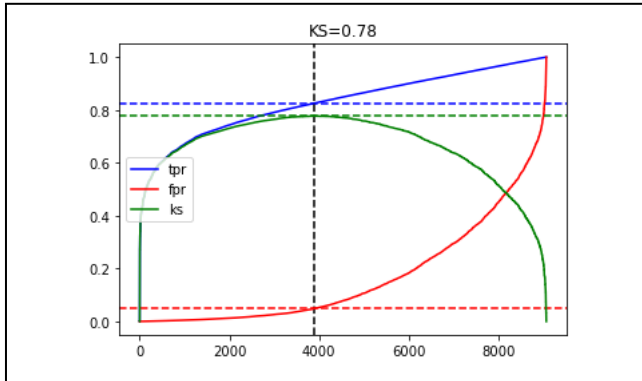


Fig. 10. Kolmogorov-Smirnov test

VIII. CONCLUSION

In this work, we outlined the concepts of credit scoring model and approaches to create efficient and effective credit scoring model. Credit scoring is used by microfinances and banks to management risk in credit lending processes such as credit cards and unsecured loan business. It is tried to measure the customer ability of willingness to pay and determine whether or not to grant credit limit. According to the experimental results, optimizing the classes weight on highly imbalanced dataset without using any oversampling techniques can also give improvement the performance of the predictive model such as logistic regression. But significantly take a long processing time can be considered as pain point and will be addressed as future work.

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